

Final Exam

Duration 1h30

All answers must be rigorously justified. Efforts should be made to ensure clear and well-structured presentation. No reference materials or calculators are allowed.

Exercise 1. (9 points)

I/ Let f be the function defined by $f(x) = \sqrt{x+6}$.

- 1. Determine the fixed points of f .*
- 2. Study the nature of the sequence defined by:*

$$\begin{cases} u_0 \geq 0, \\ u_{n+1} = f(u_n), \quad \forall n \in \mathbb{N}. \end{cases}$$

II/ Study the nature of the sequences:

$$v_n = \frac{n^3 \sin(n^2) + 3(-2)^n}{3^n}, \quad \text{and} \quad w_n = \frac{\lfloor \sqrt{n} + \cos(n) \rfloor}{2\sqrt{n} + 1}.$$

Exercise 2. (4 points)

Let $f : [0, +\infty) \rightarrow \mathbb{R}$ be the function defined by:

$$f(x) = \frac{2e^{2x}}{1 + e^x}.$$

- 1. Prove that f defines a bijection from the interval $I = [0, +\infty)$ to an interval J , which should be determined.*
- 2. Find the inverse function f^{-1} .*

Exercise 3. (5 points)

Let f be the function defined as follows:

$$f(x) = \begin{cases} \ln(1+x) - x + \frac{\pi}{4}, & \text{if } x \geq 0, \\ \arctan(ax+b) - x, & \text{if } x < 0, \end{cases}$$

where a and b are real parameters. Determine the values of a and b such that f is continuous and differentiable on $\mathbb{R}$.

Exercise 4. (2 points)

Let f be the function defined as follows:

$$f(x) = \begin{cases} \frac{\pi}{2} - \arcsin(x), & \text{if } x \in [-1, 1), \\ \sqrt{x-1}, & \text{if } x \in [1, 3]. \end{cases}$$

Can the Mean Value Theorem be applied to f on the interval $[-1, 3]$?